Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/21/2005
PI Name:	Mallis, Melissa M. Ph.D.		
Project Title:	Astronaut Scheduling Assistant: A Biomathematical Model of the Neurobehavioral Performance Capability of Astronauts		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	07/02/2001	**End Date:**	10/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronautsoften experience significant sleep loss in space. Average sleep durations are reportedly only 5-6 hrs in space compared to 7-8 hrs on the ground. This rate of sleep loss, when maintained for a one-week period, has been demonstrated in the laboratory to result in performance degradation equivalent to a single night of total sleep deprivation. Such decrements could have serious consequences for the effectiveness, health, and safety of space crews. Astronauts may presently address their space-related sleep problems via frequent use of hypnotics, but this type of pharmacological strategy is not an optimal long-term solution for the fatigue associated with space flight. A well-planned crew work/rest schedule is the key to maximizing sleep quality during bed rest and thus preserving operational performance. This project aims to develop a biomathematical model of sleep and waking neurobehavioral performance in astronauts, and to provide a software implementation of the model which can be validated in laboratory and operational environments.		

	This Astronaut Scheduling Assistant incorporates equations derived from the results of experimental studies of sleep timing and quantity, circadian rhythmicity, and waking neurobehavioral performance. Based on these equations, the model will predict the effects that acute sleep loss, cumulative sleep restriction, and circadian desynchrony may have on waking performance in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This software tool could also be used to assist in the scheduling of any ground-based crews that may encounter work hours scheduled during normal sleeping hours. This could, of course, include the mission control team.
Task Progress:	A beta version of the Astronaut Scheduling Software was tested and found to provide a functional platform for further model development. Only a minor numerical rounding problem was discovered, and subsequently fixed. The beta version did not yet include the equations for the effects of chronic sleep loss on neurobehavioral performance. These are currently being calibrated. Frequent dialogs with envisioned end users helped refine our understanding of their tasks and needs. This information is being used to drive the design of the interface, as we appreciate the importance of adhering to a user-centered design process. We believe that there simply is no other way to insure the creation of a user-friendly product. Additionally, a white paper for the requirements of the software's graphical user interface is being written on the basis of this information. We expect to validate the model predictions and operationally test the user interface in the next year.
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